Tutorial 7 report

EEE5311 (Digital IC design)

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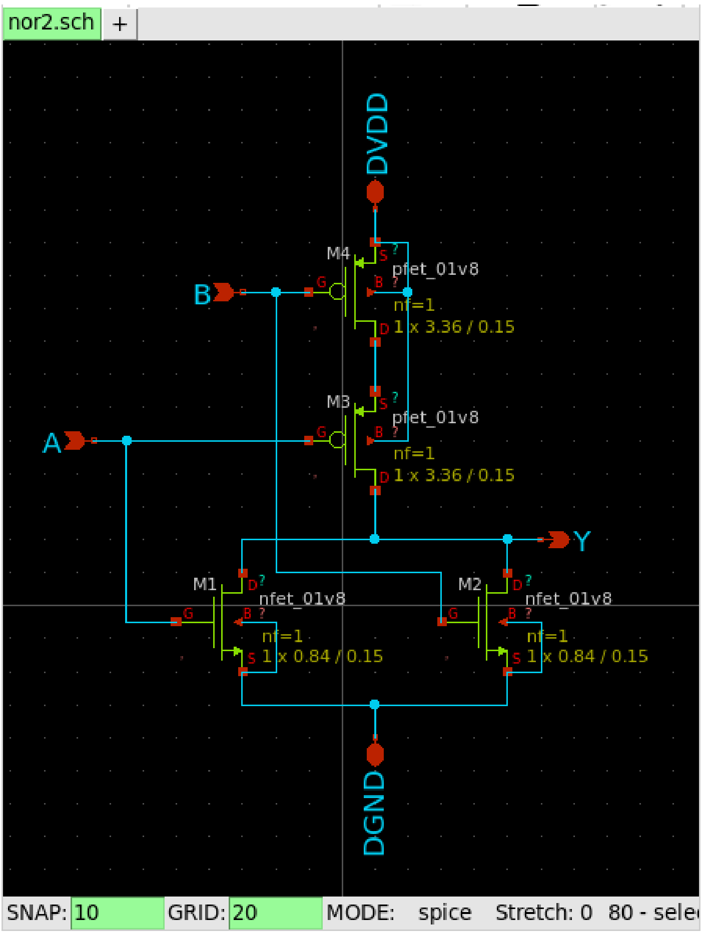
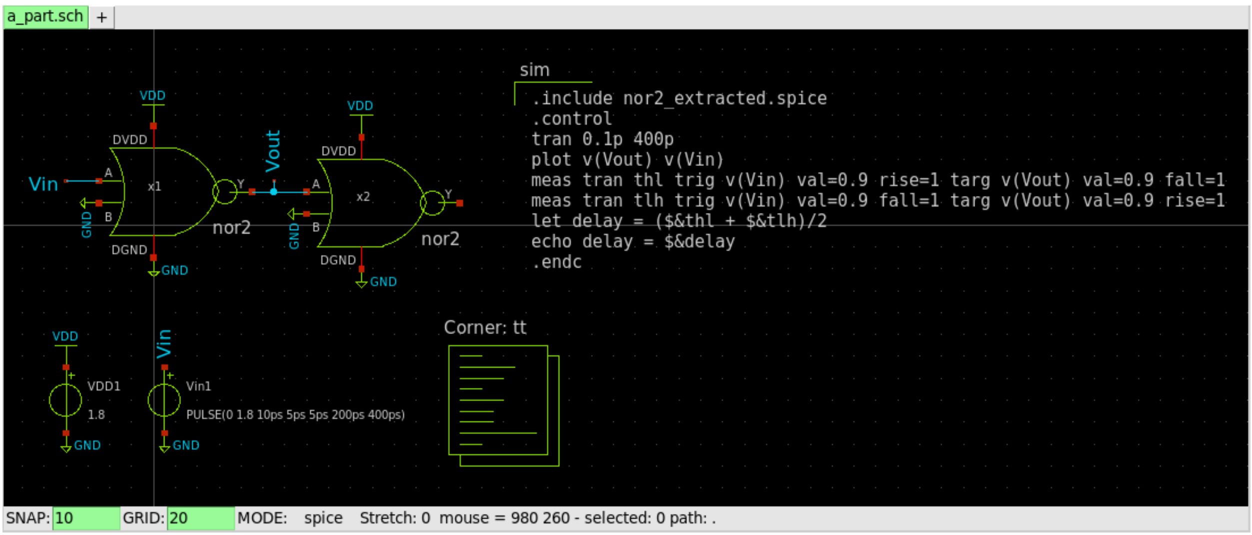
Question 1

Part a)

- Xschem schematic used for the NOR gate –



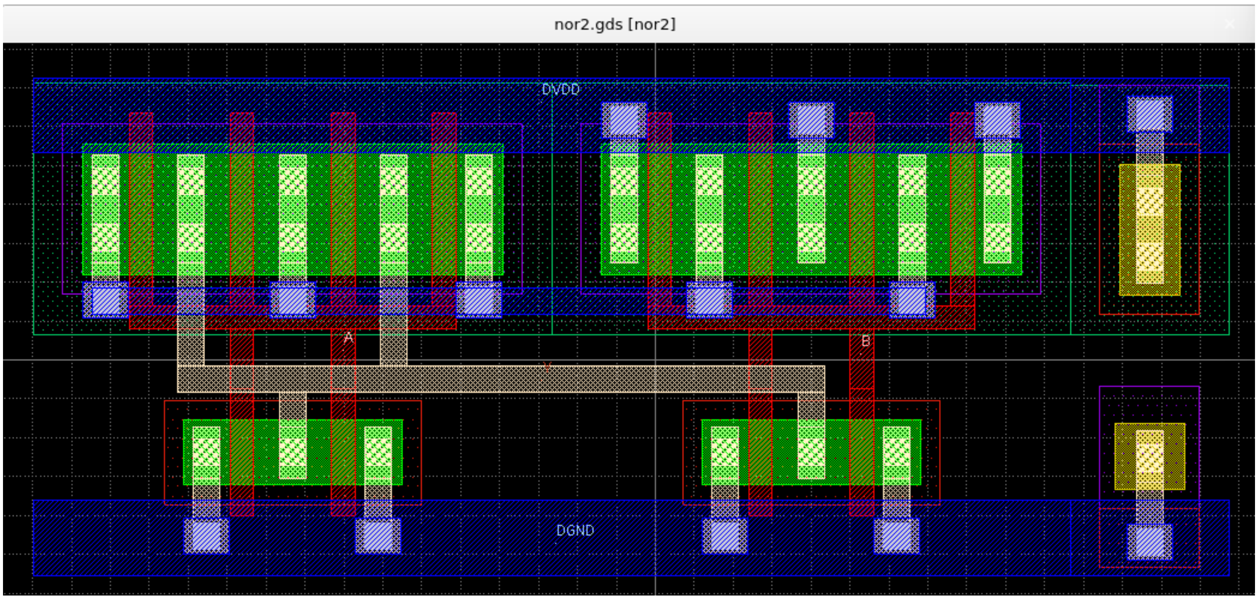
- Circuit and spice code used –



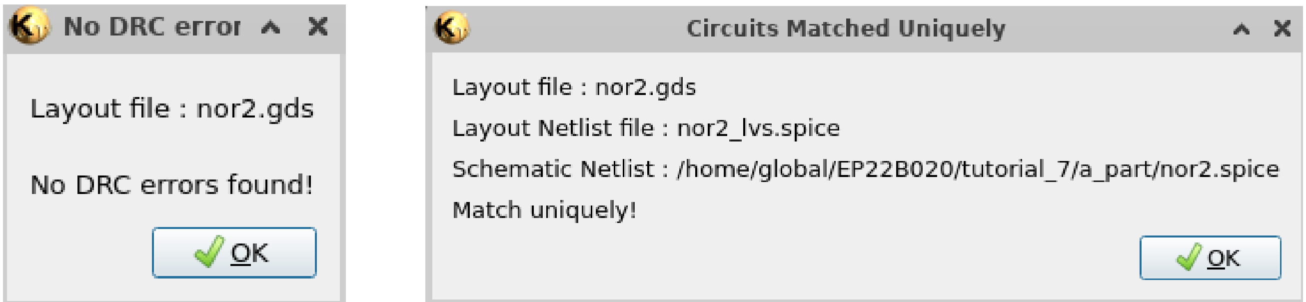
- Before extracting the layout parasitics –

```
 Reference value :  3.06350e-10
No. of Data Rows : 4020
thl                 =  2.721173e-11 targ=  3.971173e-11 trig=  1.250000e-11
tlh                 =  3.462915e-11 targ=  2.521292e-10 trig=  2.175000e-10
delay = 3.09204E-11
ngspice 16 ->
```

- Layout used in klayout –

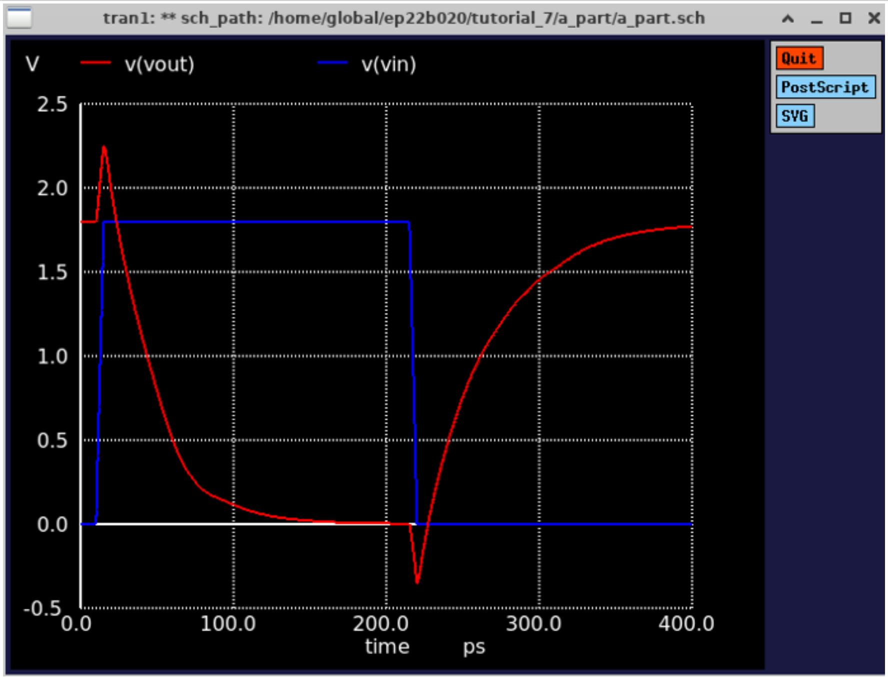


- After passing DRC and LVS –



- The delay with the layout parasitics increases –

```
 Reference value :  3.86550e-10
No. of Data Rows : 4020
thl                 =  3.407015e-11 targ=  4.657015e-11 trig=  1.250000e-11
tlh                 =  3.887205e-11 targ=  2.563721e-10 trig=  2.175000e-10
delay = 3.6471E-11
ngspice 16 ->
```



Logical effort calculations –

- For a 2-input NOR gate, g = 5/3 and p = 2. (Assuming β = 1)
- h = $(C_{self}+C_{load})/(C_{gate}) = (C_{self}+C_{gate})/(C_{gate})$ = (5+3)/3 = 8/3
- To find R_{inv} –

$$\mathrm{Rinv} = \frac{\mathrm{Req,n} + \mathrm{Req,p}}{2} = \frac{\mathrm{VDD}}{4\ln 2}\left(\frac{1}{\mathrm{IDsat,n}} + \frac{1}{\mathrm{IDsat,p}}\right)$$

$$\mathrm{IDsat,n} = 0.5 * 0.025 * 0.00834 * \frac{0.84}{0.15} * (1.8 - 0.7)^2 * \frac{0.96}{0.96 + 1.8 - 0.7} = 0.317\mathrm{mA}$$

$$\mathrm{IDsat,p} = 0.5 * 0.009 * 0.00816 * \frac{1.68}{0.15} * (1.8 - 0.7)^2 \frac{0.67}{0.67 + 1.8 - 0.7} = 0.188\mathrm{mA}$$

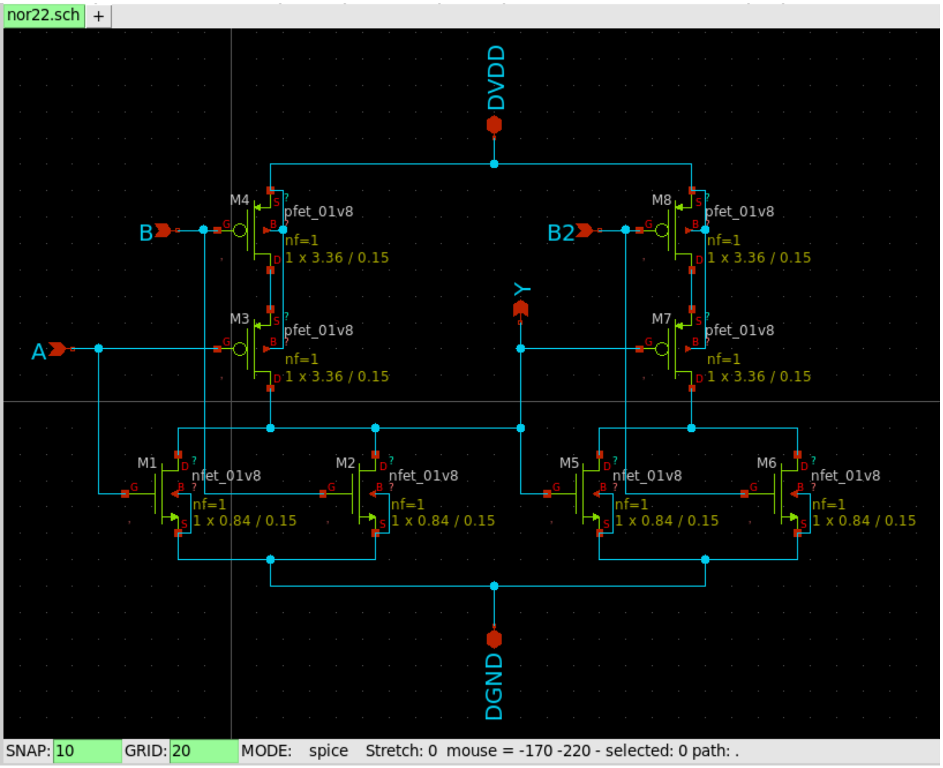
$$\mathrm{Rinv} = \frac{1.8}{2.77} * (3155.68 + 5325.25) = 5.511\mathrm{k}\Omega$$

- Cinv = 0.5*(4+2) = 3fF
- Delay = ln2*Rinv*Cinv*(p+gh) = 0.693 x 5.511 x 10^3 x 3 x 10^{-15} x (2 + 40/9)

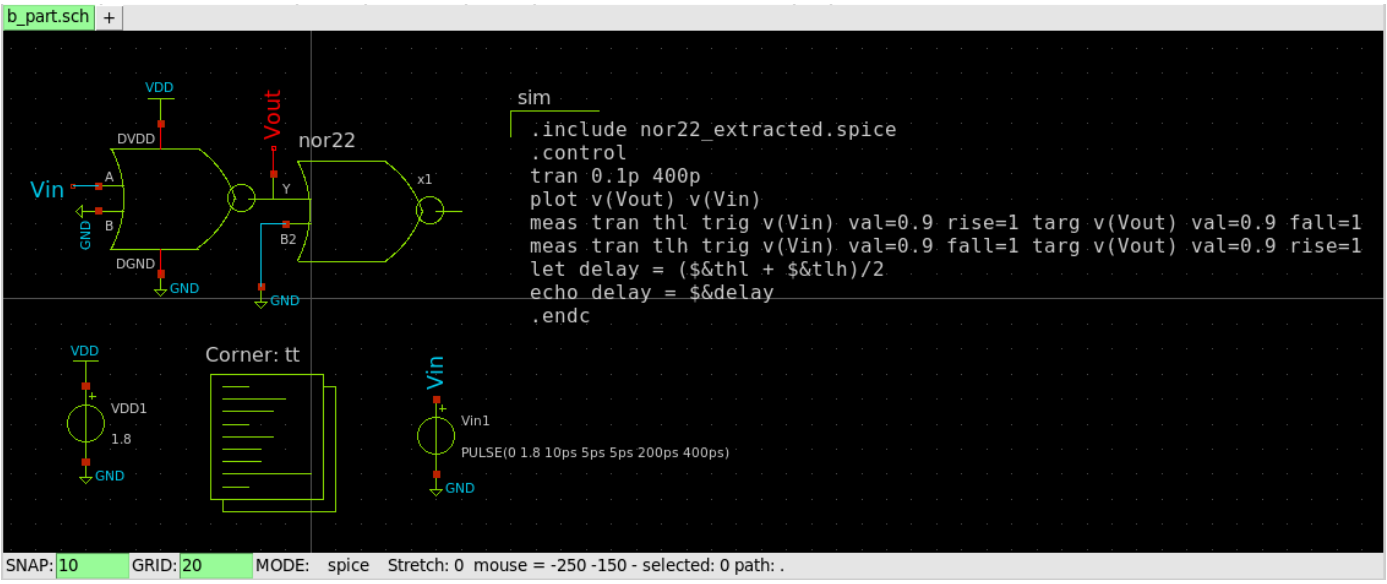
  Delay = 73852.06 x 10^{-15} s = 73.852 ps

Part b)

- Xschem schematic used for the NOR gate –



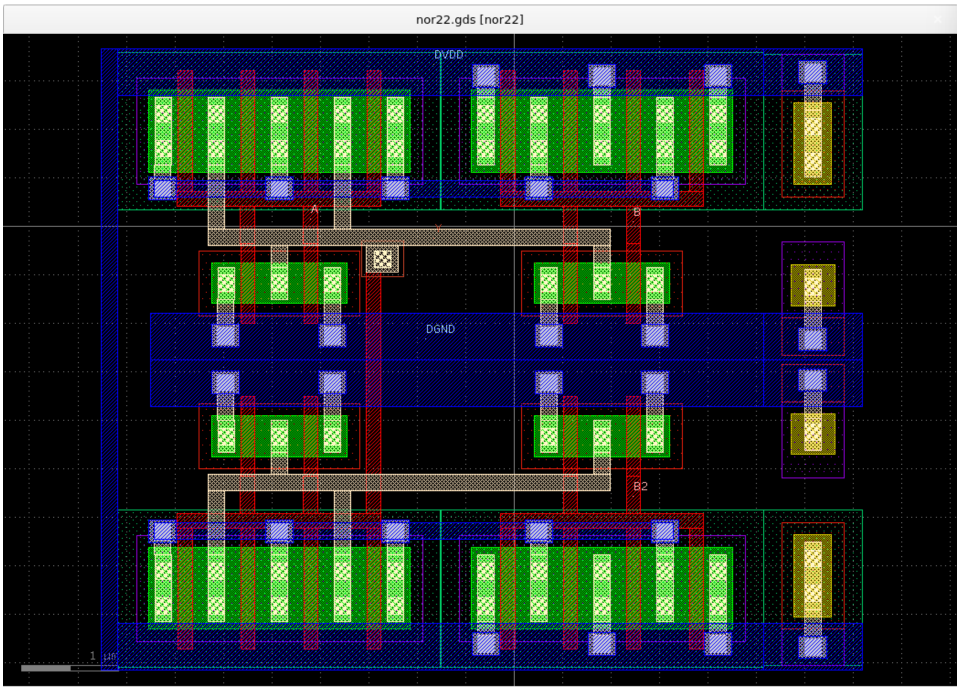
- Circuit and spice code used –



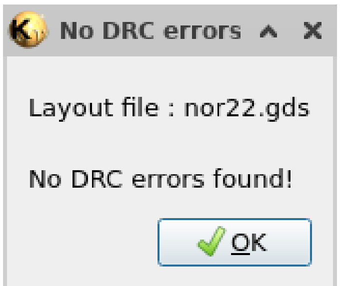
- Before extracting the layout parasitics –

```
 Reference value :  2.90650e-10
No. of Data Rows : 4020
thl                 =  2.721173e-11 targ=  3.971173e-11 trig=  1.250000e-11
tlh                 =  3.462915e-11 targ=  2.521292e-10 trig=  2.175000e-10
delay = 3.09204E-11
ngspice 16 ->
```

- Layout used in klayout –



- After passing DRC and LVS –



- The delay with the layout parasitics increases –

```
 Reference value :  3.58750e-10
No. of Data Rows : 4020
thl                  =  3.372768e-11 targ=  4.622768e-11 trig=  1.250000e-11
tlh                  =  3.753419e-11 targ=  2.550342e-10 trig=  2.175000e-10
delay = 3.5631E-11
ngspice 16 ->
```